

THE UNIVERSAL MULTIPLICITY–NORM MISMATCH, AND THE REDUCTION OF (C4) TO THE NORM-ONE SUBGROUP

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ABSTRACT. Paper XI of the series left two questions. We settle the first completely and reduce the second to a sharply posed problem, without solving it.

First question. Is there an order in a central division algebra whose number $d_{\mathfrak{p}}$ of principal right ideals of reduced norm $\mathfrak{p}$ equals $N\mathfrak{p}$, so that a boundary quotient with a nonempty KMS simplex could exist? The answer is no, and the obstruction is elementary and universal. At a prime where a maximal order in a central simple algebra of degree d has local index e , $\Lambda \otimes \mathcal{O}_{\mathfrak{p}} \cong M_{d/e}(\mathcal{O}_{D_e})$, the right ideals of reduced norm $\mathfrak{p}$ are the hyperplanes of $\mathbb{F}_{q^e}^{d/e}$, $q = N\mathfrak{p}$; hence

$$d_{\mathfrak{p}} = \frac{q^d - 1}{q^e - 1}, \quad d_{\mathfrak{p}} \geq 1 + q > q \text{ for } e < d, \quad d_{\mathfrak{p}} = 1 < q \text{ for } e = d,$$

and $d_{\mathfrak{p}} \neq N\mathfrak{p}$ at every prime, for every $d \geq 2$. For $d = 2$ the excess at a split prime is exactly 1: the off-by-one of $\mathbb{P}^1$. Eichler orders do not repair this: at a prime exactly dividing the level the local tree collapses to an edge and $d_{\mathfrak{p}} = 2$, which equals $N\mathfrak{p}$ only when $N\mathfrak{p} = 2$ — a match at one prime, never at almost all. The boundary quotient forcing the $d_{\mathfrak{p}}$ range projections to partition is in fact the zero algebra, a central element of reduced norm $\mathfrak{p}^d$ lying in every right ideal of reduced norm $\mathfrak{p}$; and even a partition in mean would force a temperature $\beta(\mathfrak{p}) = \log d_{\mathfrak{p}} / \log q$ that decreases strictly with q to $d - 1$.

Second question. Condition (C4) asks whether the measured groupoid, equivalently its Maharam extension, is non-amenable. We make two structural observations that reduce it. The Radon–Nikodym cocycle is $\text{Nrd}(a)^{-\beta}$ and therefore *factors through the reduced norm*, so the norm-one subgroup $G^1 = \ker(\text{Nrd})$ — which is non-amenable, containing free groups — acts *measure preservingly*. The non-amenable directions of G and the thermodynamic direction are orthogonal. And in the Fourier decomposition of the Maharam extension over the flow variable, the fibre representations are $\pi_{\tau} = \chi_{\tau} \otimes \pi_0$ with $\chi_{\tau}(a) = \text{Nrd}(a)^{i\beta\tau}$ a character; since $\chi_{\tau}|_{G^1} = 1$ for every τ , *all fibres restrict to one and the same representation of G^1* . The τ -integral is invisible to G^1 , and by Kuhn’s theorem (C4) follows from a single weak-containment statement: $\pi_0|_{G^1}$ not weakly contained in the regular representation of G^1 .

We do not settle it, and we say precisely why the expected arguments fail. Zimmer’s theorem that an essentially free amenable action preserving a *probability* measure forces the group to be amenable does not extend to infinite invariant measures, and the invariant measure here is infinite; and a spectral gap for $\pi_0|_{G^1}$ would not suffice, since amenable actions have tempered Koopman representations — the translation action of F_2 on itself preserves counting measure, is essentially free, has a spectral gap, and is an amenable groupoid.

1. LOCAL MULTIPLICITIES

Let K be a number field, A a central simple K -algebra of degree d , so $\dim_K A = d^2$, and $\Lambda \subset A$ an order all of whose right ideals are principal, as condition (C1) of Paper X requires; the Hurwitz order is one [4, 5]. For a finite prime $\mathfrak{p}$ of K with residue field $\mathbb{F}_q$, $q = N\mathfrak{p}$, write

$$d_{\mathfrak{p}}(\Lambda) := \#\{\text{right ideals } a\Lambda \subseteq \Lambda \text{ with } \text{Nrd}(a)\mathcal{O}_K = \mathfrak{p}\},$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XII-mismatch/](#).

the number of right ideals of reduced norm $\mathfrak{p}$, a local quantity: right ideals of Λ of reduced norm $\mathfrak{p}$ correspond to right ideals of $\Lambda \otimes \mathcal{O}_{\mathfrak{p}}$ of reduced norm $\mathfrak{p}$, by the local–global correspondence for lattices [4].

The series. The foundational paper of the series is not used here; Papers II, IX, X and XI, cited by numeral, are in preparation, and the facts taken from them are stated where used.

Lemma 1.1 (Local multiplicities). *Let Λ be maximal and let $e \mid d$ be the local index of A at $\mathfrak{p}$, so that $\Lambda \otimes \mathcal{O}_{\mathfrak{p}} \cong M_r(\mathcal{O}_D)$ with $r = d/e$ and $\mathcal{O}_D$ the maximal order of the central division algebra D of index e over $K_{\mathfrak{p}}$, whose residue field is $\mathbb{F}_{q^e}$ [4, Ch. 3]. Then*

$$(1.1) \quad d_{\mathfrak{p}} = |\mathbb{P}^{r-1}(\mathbb{F}_{q^e})| = \frac{q^d - 1}{q^e - 1} = 1 + q^e + \cdots + q^{e(r-1)}.$$

In particular $d_{\mathfrak{p}} = 1 + q + \cdots + q^{d-1}$ at a split prime ($e = 1$) and $d_{\mathfrak{p}} = 1$ at a totally ramified prime ($e = d$).

Proof. Right ideals of $M_r(\mathcal{O}_D)$ are the sets of matrices whose rows lie in a right $\mathcal{O}_D$ -submodule $L \subseteq \mathcal{O}_D^r$, and $L \mapsto$ this ideal is a bijection onto the right ideals. A generator a of the ideal has reduced norm of $\mathfrak{p}$ -adic valuation equal to the length of $\mathcal{O}_D^r/L$ as an $\mathcal{O}_D$ -module, the simple $\mathcal{O}_D$ -module being $\mathcal{O}_D/\Pi\mathcal{O}_D \cong \mathbb{F}_{q^e}$ with Π the uniformizer of D , $\text{Nrd}_D(\Pi)$ having valuation 1. So the ideals of reduced norm $\mathfrak{p}$ are the L with $\mathcal{O}_D^r/L \cong \mathbb{F}_{q^e}$, i.e. the preimages of the hyperplanes of the $\mathbb{F}_{q^e}$ -vector space $(\mathcal{O}_D/\Pi)^r = \mathbb{F}_{q^e}^r$, parametrized by the dual projective space, of cardinality (1.1). $\square$

Theorem 1.2 (Universal multiplicity–norm mismatch). *Let Λ be a maximal order in a central simple algebra of degree $d \geq 2$. Then at every prime with local index $e < d$,*

$$d_{\mathfrak{p}} - \text{N}\mathfrak{p} = 1 + q^e + \cdots + q^{e(r-1)} - q \geq 1 > 0,$$

with equality to 1 exactly when $d = 2$ and $\mathfrak{p}$ is split; and at a totally ramified prime $d_{\mathfrak{p}} = 1 < \text{N}\mathfrak{p}$. In particular $d_{\mathfrak{p}} \neq \text{N}\mathfrak{p}$ for every $d \geq 2$ and every $\mathfrak{p}$, and there is no maximal order in a central simple algebra for which the multiplicity matches the norm at even one prime.

Proof. By Lemma 1.1, for $e < d$ we have $r \geq 2$ and $d_{\mathfrak{p}} - q = 1 + q^e + \cdots + q^{e(r-1)} - q$; if $e = 1$ and $d = 2$ this is $1 + q - q = 1$; if $e = 1$ and $d \geq 3$ it is $1 + q^2 + \cdots + q^{d-1} \geq 1 + q^2 > 1$; if $e \geq 2$ it is at least $1 + q^e - q > 1$. For $e = d$, $d_{\mathfrak{p}} = 1 < q$. $\square$

Proposition 1.3 (Eichler and hereditary orders). *Let $d = 2$ and let Λ be an Eichler order of level $\mathfrak{N}$. At a prime $\mathfrak{p} \parallel \mathfrak{N}$ the local order is conjugate to $\begin{pmatrix} \mathcal{O}_{\mathfrak{p}} & \mathcal{O}_{\mathfrak{p}} \\ \mathfrak{p} & \mathcal{O}_{\mathfrak{p}} \end{pmatrix}$, the local tree collapses to a single edge, and $d_{\mathfrak{p}} = 2$. Hence $d_{\mathfrak{p}} = \text{N}\mathfrak{p}$ if and only if $\text{N}\mathfrak{p} = 2$. At the primes not dividing $\mathfrak{N}$ the order is maximal and Theorem 1.2 applies. Consequently $d_{\mathfrak{p}} = \text{N}\mathfrak{p}$ can hold at finitely many primes at most, and never at almost all primes.*

Proof. The local Eichler order of level $\mathfrak{p}$ is the stabilizer of an edge of the Bruhat–Tits tree, whose two endpoints give the two right ideals of reduced norm $\mathfrak{p}$; the remaining $q - 1$ neighbours of each vertex are not stabilized. The level is divisible by only finitely many primes. $\square$

2. THE BOUNDARY QUOTIENT IS EMPTY, UNIVERSALLY

Theorem 2.1 (The boundary quotient is zero). *Let Λ be an order in a central simple algebra of degree $d \geq 2$ over K , all of whose right ideals are principal, $P = \Lambda \setminus \{0\}$, and let $\mathfrak{p} = (\pi)$ be a principal prime of K . Let A be a unital C^* -algebra generated by isometries v_a , $a \in P$, with $v_a v_b = v_{ab}$. If two range projections e_{a_j}, e_{a_k} of distinct right ideals of reduced norm $\mathfrak{p}$ are orthogonal in A , then $A = 0$. In particular the boundary quotient of the Nica–Toeplitz algebra $\mathcal{T}(P)$ by the relations $\sum_{k=1}^{d_{\mathfrak{p}}} e_{a_k} = 1$, at any one such prime, is the zero algebra and carries no state.*

Proof. For $a \in \Lambda$ the reduced characteristic polynomial $t^d - c_1 t^{d-1} + \dots \pm \text{Nrd}(a)$ has coefficients in $\mathcal{O}_K$ and vanishes at a , so $\text{Nrd}(a) = a b$ with $b = \pm(a^{d-1} - c_1 a^{d-2} + \dots) \in \Lambda$ [4, §9]: the central element $\text{Nrd}(a)$ lies in $a\Lambda$. For a_j of reduced norm $\mathfrak{p} = (\pi)$ we have $\text{Nrd}(a_j) = \pi u_j$ with $u_j \in \mathcal{O}_K^\times$, so $\pi = a_j b_j u_j^{-1} \in a_j \Lambda$ and $v_\pi = v_{a_j} v_{b_j} u_j^{-1}$, whence $e_\pi = v_\pi v_\pi^* \leq v_{a_j} v_{a_j}^* = e_{a_j}$; likewise $e_\pi \leq e_{a_k}$. If $e_{a_j} e_{a_k} = 0$ then $e_\pi = 0$, so $v_\pi = e_\pi v_\pi = 0$ and $1 = v_\pi^* v_\pi = 0$. Finally a sum of projections equal to 1 is an orthogonal sum, as in Paper XI: $e_j = e_j \sum_k e_k$ gives $\sum_{k \neq j} e_j e_k e_j = 0$ with positive summands, so $e_j e_k = 0$. $\square$

Remark 2.2 (Partition in mean). Even the weaker requirement $\sum_{k=1}^{d_{\mathfrak{p}}} \varphi(e_{a_k}) = 1$ for a KMS_β state φ of $\mathcal{T}(P)$ itself, with norm $N = |\text{Nrd}|^t$, forces $d_{\mathfrak{p}} q^{-t\beta} = 1$, i.e.

$$\beta = \beta(\mathfrak{p}) = \frac{\log d_{\mathfrak{p}}}{t \log q} = \frac{1}{t} \cdot \frac{\log((q^d - 1)/(q^e - 1))}{\log q}$$

at a prime of local index e . For split primes the function $q \mapsto \log((q^d - 1)/(q - 1))/\log q$ is strictly decreasing with limit $d - 1$ as $q \rightarrow \infty$ and value $> d - 1$ for every finite q ; so distinct residue characteristics give distinct $\beta(\mathfrak{p})$, and no single β serves two split primes. Numerically, for $d = 2$: $\beta(\mathfrak{p})t = 1.5850, 1.2619, 1.1133, 1.0289, 1.0021$ at $q = 2, 3, 5, 13, 101$, decreasing to 1; for $d = 3$: $2.8074, 2.3347, 2.1337, 2.0310, 2.0022$, decreasing to 2; for $d = 4$, decreasing to 3.

Remark 2.3 (Why the affine systems are different). In the $ax + b$ systems of Paper II the multiplicity at $\mathfrak{p}$ is the number $\text{N}\mathfrak{p}$ of additive cosets, exactly equal to the norm, and the translates of a fundamental domain are disjoint, so the relation reads $\text{N}\mathfrak{p} \cdot \text{N}\mathfrak{p}^{-\beta} = 1$ and forces $\beta = 1$ *independently of $\mathfrak{p}$* : a KMS state survives at that one temperature. The matching there is not an accident but the statement that $\mathcal{O}_K/\mathfrak{p}$ has $\text{N}\mathfrak{p}$ elements. Theorem 1.2 says that the multiplicative analogue — counting ideals rather than cosets — can never match, the projective space $\mathbb{P}^{d-1}(\mathbb{F}_q)$ having one point too many.

Corollary 2.4 (The dichotomy of Paper XI cannot be broken). *The question of Paper XI — whether some order has $d_{\mathfrak{p}} = \text{N}\mathfrak{p}$, so that a boundary quotient with nonempty KMS simplex could exist — is answered in the negative by Theorem 1.2 and Proposition 1.3, and Theorem 2.1 shows that the multiplicity is not even the point: the quotient is zero whatever $d_{\mathfrak{p}}$ is. The nested/partition dichotomy must therefore be circumvented rather than broken, which means working with the Toeplitz algebra and its Maharam extension.*

3. THE COCYCLE FACTORS THROUGH THE REDUCED NORM

We turn to (C4), for the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$. Let $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ be the group of fractions, $\tilde{\Omega}$ the dilation carrying the G -action with the measure $\tilde{\nu}_\beta$ extending a KMS_β measure, $\tilde{\nu}_\beta(aE) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$, and

$$G \curvearrowright (\tilde{\Omega} \times \mathbb{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (ax, s + \beta \log \text{Nrd}(a))$$

the Maharam extension, which preserves the displayed measure: the pushforward of $\tilde{\nu}_\beta \otimes e^s ds$ under a scales the first factor by $\text{Nrd}(a)^{-\beta}$ and the second by $e^{\beta \log \text{Nrd}(a)}$. By Takesaki duality, as in Paper XI, (C4) — non-amenability of the measured groupoid of G on $(\tilde{\Omega}, \tilde{\nu}_\beta)$, equivalently non-injectivity of the factor — is equivalent to non-amenability of the Maharam extension. Put

$$G^1 := \ker(\text{Nrd} : G \rightarrow \mathbb{Q}_+^\times).$$

Proposition 3.1 (Orthogonality of the two directions). (1) *The Radon–Nikodym cocycle of the G -action on $(\tilde{\Omega}, \tilde{\nu}_\beta)$ is $a \mapsto \text{Nrd}(a)^{-\beta}$, which factors through Nrd . Consequently G^1 acts measure preservingly, and acts trivially on the flow variable s .*

- (2) $G^1 = \mathbb{H}(\mathbb{Q})^1$, the norm-one rational quaternions, is non-amenable: it contains free subgroups of rank five.
- (3) Hence the modular direction of the system, which is the whole content of the dynamics σ_t , lies in the abelian quotient $G/G^1 \cong \text{Nrd}(G) \leq \mathbb{Q}_+^\times$, while all the non-amenableity of G lies in G^1 , on which the dynamics is trivial.

Proof. (1) The scaling relation $\tilde{\nu}_\beta(aE) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$ depends on a only through $\text{Nrd}(a)$; for $a \in G^1$ it is the identity, and $s \mapsto s + \beta \log \text{Nrd}(a) = s$. (2) By Lubotzky–Phillips–Sarnak [3, §3] the quaternions $1 \pm 2i, 1 \pm 2j, 1 \pm 2k$ of reduced norm 5 generate, modulo the centre, a free group $\Lambda(5)$ of rank three; the elements of $\Lambda(5)$ of reduced norm 5^{2m} form a subgroup of index two, free of rank five, and $\alpha \mapsto \alpha/5^m$ embeds it into $\mathbb{H}(\mathbb{Q})^1/\{\pm 1\}$. (3) is (1) and (2). $\square$

Remark 3.2 (What this changes). Proposition 3.1 is the reason the Hurwitz system is not merely a non-commutative decoration of the abelian one. In every earlier model the acting group was abelian and its non-amenableity was not merely absent but impossible. Here the group splits cleanly: a non-amenable kernel acting by measure-preserving transformations, and an abelian quotient carrying the entire thermodynamics. If (C4) holds it will be because of G^1 , and G^1 is invisible to σ_t .

4. ALL FOURIER FIBRES AGREE ON G^1

Proposition 4.1 (Fibre decomposition). *Conjugating by $\xi \mapsto e^{s/2}\xi$ identifies $L^2(\mathbb{R}, e^s ds)$ with $L^2(\mathbb{R}, ds)$, and carries the Koopman representation of the Maharam extension to $(\pi(a)F)(x, s) = \text{Nrd}(a)^{\beta/2} F(a^{-1}x, s - \beta \log \text{Nrd}(a))$; Fourier transform in s then gives*

$$L^2(\tilde{\Omega} \times \mathbb{R}, \tilde{\nu}_\beta \otimes e^s ds) \cong \int_{\mathbb{R}}^{\oplus} \mathcal{H}_\tau \frac{d\tau}{2\pi}, \quad \mathcal{H}_\tau = L^2(\tilde{\Omega}, \tilde{\nu}_\beta),$$

with fibre representations

$$(4.1) \quad \pi_\tau(a) = \chi_\tau(a) \pi_0(a), \quad \chi_\tau(a) = \text{Nrd}(a)^{i\beta\tau}, \quad (\pi_0(a)f)(x) = \text{Nrd}(a)^{\beta/2} f(a^{-1}x).$$

Here π_0 is a genuine unitary representation of G on $L^2(\tilde{\Omega})$, the dilation being what makes the isometries of P into unitaries.

Proof. Standard: π_0 is the Koopman representation of the non-singular action, with Radon–Nikodym factor $\text{Nrd}(a)^{\beta/2}$, and the translation by $\beta \log \text{Nrd}(a)$ in s becomes multiplication by $\chi_\tau(a)$ on the τ -th Fourier component, the sign of the exponent being a convention. The only point to note is that on Ω itself $\pi_0(a)$ is an isometry with range $L^2(a\Omega) \subsetneq L^2(\Omega)$, and becomes unitary only after passing to the dilation $\tilde{\Omega} = \varinjlim (\Omega, a \cdot)$. $\square$

Theorem 4.2 (The τ -decomposition is invisible to G^1). *For every $\tau \in \mathbb{R}$, $\chi_\tau|_{G^1} = 1$, hence*

$$\pi_\tau|_{G^1} = \pi_0|_{G^1} \quad \text{for all } \tau.$$

Consequently the restriction to G^1 of the Koopman representation of the Maharam extension is $\pi_0|_{G^1} \otimes L^2(\mathbb{R})$, a multiple of a single representation, and

$$(4.2) \quad (C4) \iff \pi_0|_{G^1} \text{ is not weakly contained in the regular representation } \lambda_{G^1}.$$

Proof. $\chi_\tau(a) = \text{Nrd}(a)^{i\beta\tau} = 1$ for $a \in G^1$. The displayed restriction follows from (4.1) and Proposition 4.1. For (4.2): if the action of G on $(\tilde{\Omega}, \tilde{\nu}_\beta)$ were amenable, so would be its restriction to the subgroup G^1 [1]; and by Kuhn’s theorem [2] the Koopman representation of an amenable action is weakly contained in the regular representation, so $\pi_0|_{G^1} \prec \lambda_{G^1}$. $\square$

Remark 4.3 (A spectral gap would not suffice). The converse of Kuhn's theorem fails, so the implication (4.2) cannot be weakened to the absence of almost invariant vectors: the translation action of F_2 on itself is amenable, being proper, and its Koopman representation is λ_{F_2} , which has a spectral gap. A gap for $\pi_0|_{G^1}$ is necessary for (4.2), since λ_{G^1} has one, but it is the failure of temperedness that is needed.

Remark 4.4. Theorem 4.2 removes the direct integral from the problem. Whatever is to be established about weak containment, it concerns one representation of one group — the measure-preserving Koopman representation of the norm-one subgroup — and not a family of questions indexed by τ . This is the concrete simplification we can offer.

5. WHY THIS DOES NOT YET CLOSE (C4)

Remark 5.1 (The failure of the expected argument). The natural next step is Zimmer's theorem: an essentially free amenable action of Γ preserving a probability measure forces Γ to be amenable. Applied to $G^1 \curvearrowright (\tilde{\Omega}, \tilde{\nu}_\beta)$ with G^1 non-amenable it would give (C4) at once. It does not apply, for a specific reason: $\tilde{\nu}_\beta$ is *infinite*. Indeed $\tilde{\nu}_\beta(a^{-1}\Omega) = \text{Nrd}(a)^\beta \rightarrow \infty$, so the dilation carries only a σ -finite invariant measure.

That the extension from finite to infinite invariant measure genuinely fails is shown by an example: the translation action of F_2 on itself preserves counting measure, is essentially free, and its orbit groupoid is transitive with trivial isotropy, hence proper, hence amenable. Non-amenability of the group is therefore not transmitted by an infinite-measure-preserving essentially free action, and by Remark 4.3 not by a spectral gap either.

Remark 5.2 (What would suffice). Two routes seem available, and we have carried out neither.

- (1) Find a G^1 -invariant subset of finite measure meeting almost every orbit, or more generally show that the restricted relation $\mathcal{R}_{G^1}|_\Omega$ on the probability space (Ω, ν_β) is of type II₁ and is the orbit relation of an essentially free pmp action of a non-amenable group. The difficulty is that Ω is not G^1 -invariant, so the restricted relation is not literally an orbit relation.
- (2) Show directly that $\pi_0|_{G^1}$ is not weakly contained in λ_{G^1} — for instance by exhibiting a finite-dimensional subrepresentation, which an infinite-measure Koopman representation need not have. By Theorem 4.2 this is one representation; by Proposition 3.1 it is measure preserving; and G^1 contains the LPS free groups, for which spectral gap statements are available on *congruence quotients* [3]. Whether any of that transfers to $\tilde{\Omega}$, which is not a congruence quotient, we do not know — Paper XI warns that the tree methods behind those statements are unavailable here, and Remark 4.3 that a gap alone would in any case not be enough.

Remark 5.3 (Essential freeness). Route (i) needs the G^1 -action to be essentially free, which we have not verified. The isotropy of the G -action on $\tilde{\Omega}$ was computed in the abelian case in Paper II and found to be trivial off a set that the KMS measures do not charge; the analogous computation here would have to handle the 24 units and the non-commutativity, and we have not done it. Zimmer's theorem is used here in the form of [6, Ch. 4].

6. ASSESSMENT

Question	Answer	Reference
d_p at a prime of local index e , maximal order	$(q^d - 1)/(q^e - 1)$	Lem. 1.1
$\exists$ order with $d_p = Np$?	no, for every $d \geq 2$	Thm. 1.2
excess $d_p - Np$ for $d = 2$	exactly 1	Thm. 1.2
Eichler orders	$d_p = 2$ at $p \parallel \mathfrak{N}$; $= Np$ iff $Np = 2$	Prop. 1.3
boundary quotient nonzero?	never	Thm. 2.1
RN cocycle factors through Nrd	yes	Prop. 3.1
G^1 acts measure-preservingly	yes, and is non-amenable	Prop. 3.1
Fourier fibres on G^1	all equal to $\pi_0 _{G^1}$	Thm. 4.2
$(C4) \Leftarrow \pi_0 _{G^1} \not\Leftarrow \lambda_{G^1}$	true (Kuhn)	Thm. 4.2
Zimmer closes (C4)	no: invariant measure infinite	Rem. 5.1
a spectral gap closes (C4)	no: amenable actions are tempered	Rem. 4.3
(C4)	open	—

Task 1 is closed. The multiplicity d_p counts points of a projective space and the norm Np counts points of an affine one; the difference is the hyperplane at infinity, and it is never zero. The vanishing of the boundary quotient found in Paper XI for the Hurwitz order is therefore a universal feature of orders in central simple algebras, and the nested/partition dichotomy has to be circumvented rather than broken.

Task 2 is not closed, but it is now a single question rather than a programme. The decomposition of G given by Proposition 3.1 — non-amenable kernel acting measure-preservingly, abelian quotient carrying the dynamics — is the structural feature that distinguishes this system from every earlier one, and Theorem 4.2 reduces the analysis to one representation of that kernel. What is missing is a finite-measure foothold together with essential freeness, or a proof that $\pi_0|_{G^1}$ is not tempered. If $\pi_0|_{G^1}$ is not weakly contained in the regular representation, the Hurwitz factor is the first non-injective arithmetic Bost–Connes factor and the deformation/rigidity machinery that Paper IX found inapplicable becomes actionable; if the action is amenable, the factor collapses like all the others and the programme reaches its end.

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